BABU SOVEREIGN PERUMALLA

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SUMMARY

Senior technical lead with 20 years building production systems where failure is expensive: payment gateway and point-of-sale services for Safeway, real-time store metrics for ~2,200 Albertsons stores, and commerce platforms for Macy's, Gap and Williams-Sonoma. Since 2025, building LLM safety evaluations on my own time, focused on adversarial robustness and on how far an automated judge can be trusted. Moving into AI control and oversight work.

INDEPENDENT AI SAFETY AND EVALUATION WORK

Independent Projects 2025 – Present

- **Eval harness:** Built a local prompt-coaching tool with a 143-case evaluation harness across six suites (adversarial, kid safety, false positive, urgent, normal, feedback): 6 runs per case at mixed temperatures, an LLM judge, release gates, canary-based instruction-bypass tests and regression baselines; ~58k lines of Python with property-based and mutation testing. The harness currently blocks its own release (1 of 29 adversarial cases pass).
- **Finding:** In a controlled swap of only the optimizer model, Claude vs. qwen2.5-7B raised fact retention from 70% to 96% and urgent-suite pass rate from 50% to 100%, but adversarial soft failures only fell from 80% to 72%; the weaker judge (gemma3:27b) scored the stronger model lower on meaning preservation (81% vs. 98%).
- **Judge audit:** Running a blind 80-item human audit of the LLM judge to measure where a weaker overseer misjudges a stronger model.
- **Builds:** SkyWatch, a Claude agent that identifies 10,000+ live flights from the web (built in one evening at a Claude Code Build Night, SF); Robin Hoodwinked, a logic game about spotting liars (Claude Community Build Day, SF); Morsel Jar, an on-device iOS app (App Store).

PROFESSIONAL EXPERIENCE

Nisum | Technical Lead II (Senior Technical Lead), client engagements for national US retailers 2006 – Present

Williams-Sonoma | Search and Product Discovery (Denali) Current

- Tech lead for backend services behind shop and search across multiple brands: AI Related Searches, Popular Filters, catalog and product-image feeds, comparison tools, Inspiration Board user-generated content, and customer profile services.
- Own technical designs and sizing; coordinate onsite and offshore engineers. Java, Spring Boot, Cassandra, S3, Kubernetes.

Albertsons Companies | Store Mobility (SOSM)

- Architected the store-order metrics pipeline: order-status events from multiple Kafka sources, streaming metric jobs in Scala and PySpark on Databricks (GCP), aggregated store → district → division, and served over GraphQL to the iOS app used in ~2,200 stores.

Safeway (Albertsons Companies) | Payments: VPOS, ECHO, Soteria

- Subject-matter expert for VPOS, the point-of-sale service for purchases, refunds and receipts in Safeway stores nationwide; owned payment operations and receipt processing end to end, including ECHO refund and account-credit tools for support agents.
- Built REST services for Soteria, the payment gateway for web and POS, which tokenized the online purchase path through a third-party vault so no service held a real card number.

Gap Inc. | Demand Forecasting (VMI / PETE) and PIM Photography

- Designed the architecture for the PETE demand-forecast extract flows that feed vendor replenishment; set up the PCF, Jenkins, FedCI and OpenStack delivery pipeline. Also led backend work on the PIM photography workflow (shot requests, crop validation).

Backcountry | Fulfillment

- Migrated warehouse and order management (WMS/OMS) workflows for inventory, orders and shipments from legacy systems to GCP microservices (Java, Kotlin, MongoDB), including admin and handheld picking and packing tools.

Macy's | Team Lead, then Senior Technical Lead (2013 – 2016) 2009 – 2016

- Led 20 engineers on iPOP (Improved Presentation of Product), which targeted $165M in incremental sales.
- Led teams of 8–10 on Colorway Pricing, Continuous Delivery, and Deals & Promotions migration.
- Holiday Stability 2012: migrated a datastore under live peak-season traffic.

SKILLS

Languages and backend: Java, Python, Scala, Kotlin, SQL, Bash; Spring Boot, Spring Core/MVC, Hibernate, REST, GraphQL, SOAP
Data and streaming: Kafka, Databricks, PySpark, Cassandra, MongoDB, Oracle, DB2, MySQL, SQL Server, Snowflake
Cloud and delivery: GCP, Kubernetes (AKS), Amazon S3, Pivotal Cloud Foundry, Jenkins, GitHub Actions, Maven, Gradle
AI: Claude API, Model Context Protocol (MCP), Claude Code, Ollama, LLM-as-judge evaluation, GitHub Copilot, Codex

EDUCATION AND TRAINING

B.Tech, Computer Science and Engineering | Vignan Institute of Information Technology (JNTUH), India 2006

Machine Learning Specialization, Stanford / DeepLearning.AI (3 courses, 2025–2026); Anthropic courses on Claude Code, subagents, agent skills and MCP (2026). Topcoder US Ambassador and talk on AI developer tools (2024).